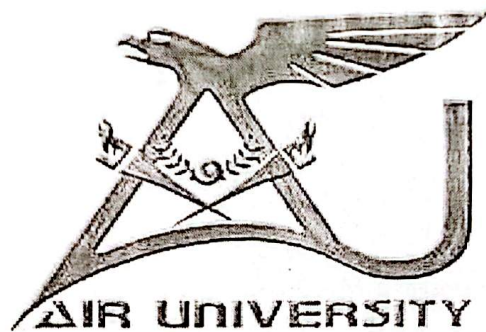


Mechanics of Machines

Project Report

Title: Umbrella Opening/Closing Mechanism



Class: BEME-F-24-B

Group Members:

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Date: June 9, 2026

Introduction: Shade and Shelter are essential for human comfort especially in regions with intense sunlight, and large retractable umbrella systems provide an elegant solution for outdoor public spaces and commercial environments. The Umbrellas in Masjid al-Nabwi, that open and close automatically to protect pilgrims from the sun, and our mechanism replicates the same fundamental working principal on a smaller, study-project scale. The mechanism convert the rotary motion of an electric motor into the linear reciprocating motion of a slider which drives the rocker arms upward or downward to open or close the umbrella canopy.

Components of Umbrella:

- Rocker arms (4)
- Connecting rods (pinned with sliders) (5)
- Slider (1)
- Crank (1)
- Pins (5-10)
- Central Column (1)

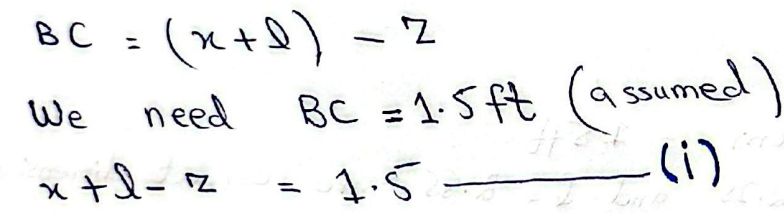
Material Used:

- Mild Steel (ASTM A36)
- Aluminum Alloy (only for Rocker Arms)
Slider and coupler

Assumptions:

- Umbrella closing/opening time is 30sec.
- Slider velocity is constant
- All components are assumed to be rigid bodies, elastic deformation is neglected in kinematic analysis.
- Wind load on the canopy is not considered in this basic design.
- The slider moves purely

- position)



We need $BC = 1.5 \text{ ft}$ (assumed)

$$\Delta OAB$$

$$x^2 + z^2 = l^2 \quad \text{--- (ii)}$$

Checking 1 and 2 for all values of x .

Selecting this
value for x ,
 z and l

Slider position =

3cm

[Extreme position]

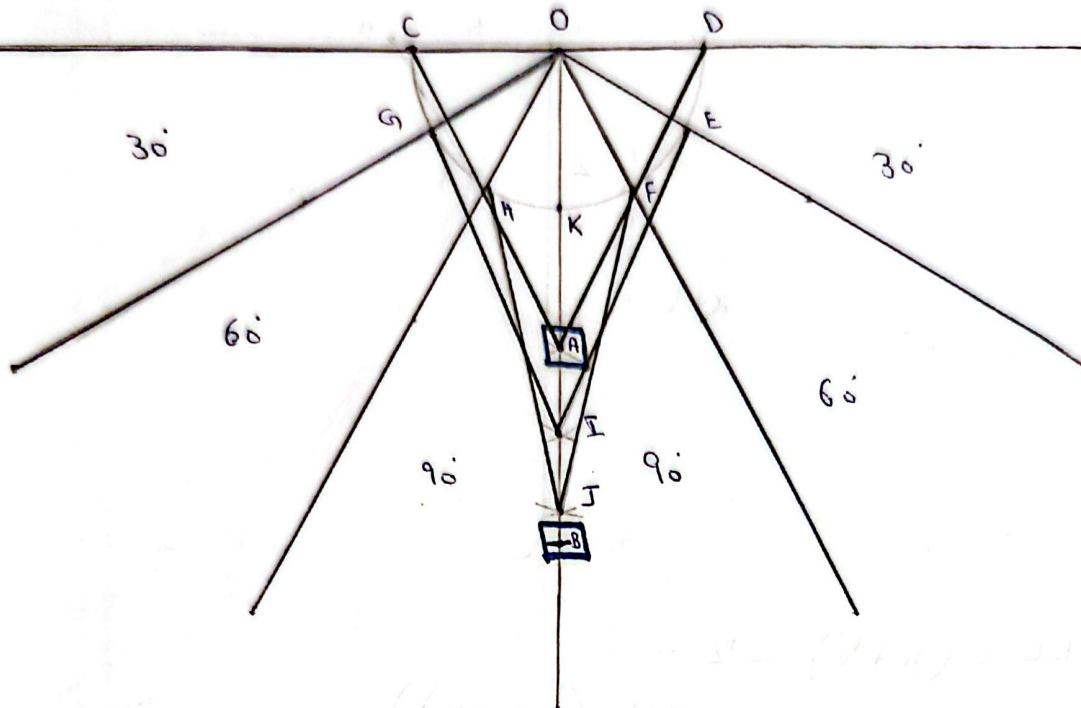
Scaling up (2) times

$$x = 1.2 \times 2 = 2.4 \text{ cm}$$

$$z = 2.25 \times 2 = 4.5 \text{ cm}$$

$$l = 2.55 \times 2 = 5.1 \text{ cm}$$

$$1 \text{ ft} = 2 \text{ cm}$$



→ Slider Stroke = 3cm = 1.5ft

So, $x = 1.2$, $z = 2.25$ and $l = 2.55$ are correct dimensions

→ z is extreme position of slider where the umbrella fully opens. (Point A)

→ $x + l = 1.2 + 2.55 = 3.75 \text{ ft}$
mean position where the umbrella fully closed.

→ Now, doing these:

- Position analysis
- Velocity analysis
- Acceleration analysis

Position Analysis

Given that: Rocker angle = 0°

Coupler / connector angle = 28°

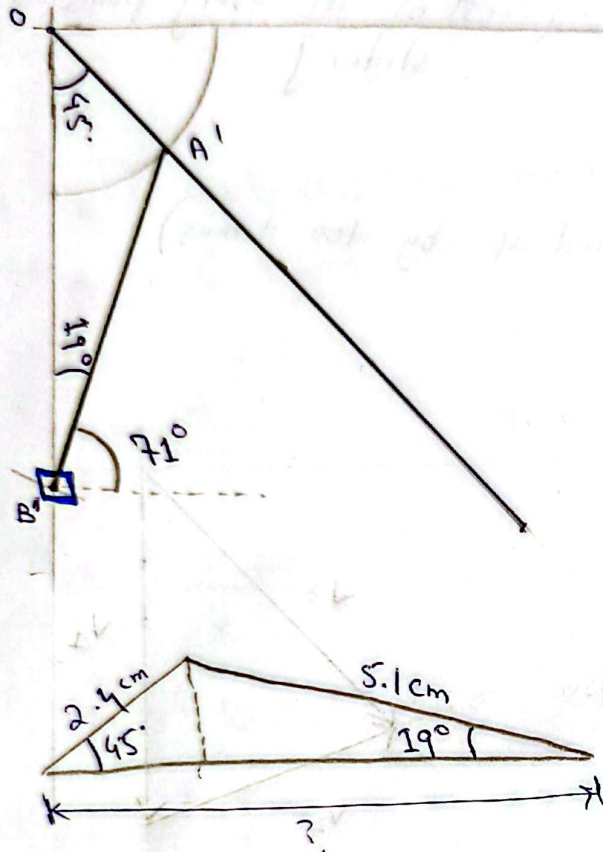


Slider position = 3cm [Extreme position]

When Rocker = -45° or 315° from x -axis

Coupler / Connector angle = ?

Slider position = ?



Coupler angle = 71°

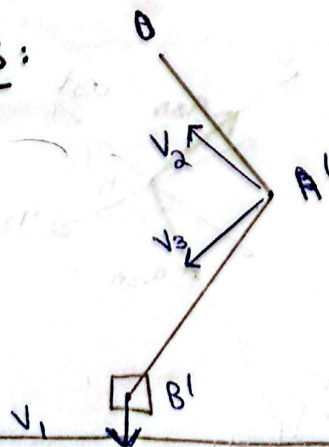
Slider position:

$$S.P = (x+2) - (2.4 \cos 45 + 5.1 \cos 19)$$

$$S.P = (2.4 + 5.1) - (1.697 + 5.1)$$

$$S.P = 0.981 \text{ cm (from mean position)}$$

Velocity analysis:



Slider velocity:

$$V_1 = \frac{\text{Slider Stroke}}{\text{total time for closing umbrella}}$$

$$V_1 = \frac{1.5 \text{ ft}}{30 \text{ sec}}$$

$$V_1 = 0.05 \text{ ft/sec} \quad [\text{constant at every point of slider}]$$

Velocity Diagram:

$$V_1 = 0.05 \Rightarrow 5 \text{ cm} \quad (\text{scaled up by 100 times})$$

$$V_2 = 3.9 \text{ cm} = 0.039 \text{ ft/sec}$$

$$V_3 = 6.25 \text{ cm} = 0.0625 \text{ ft/sec}$$

$$\omega_2 = \frac{V_2}{r_2}$$

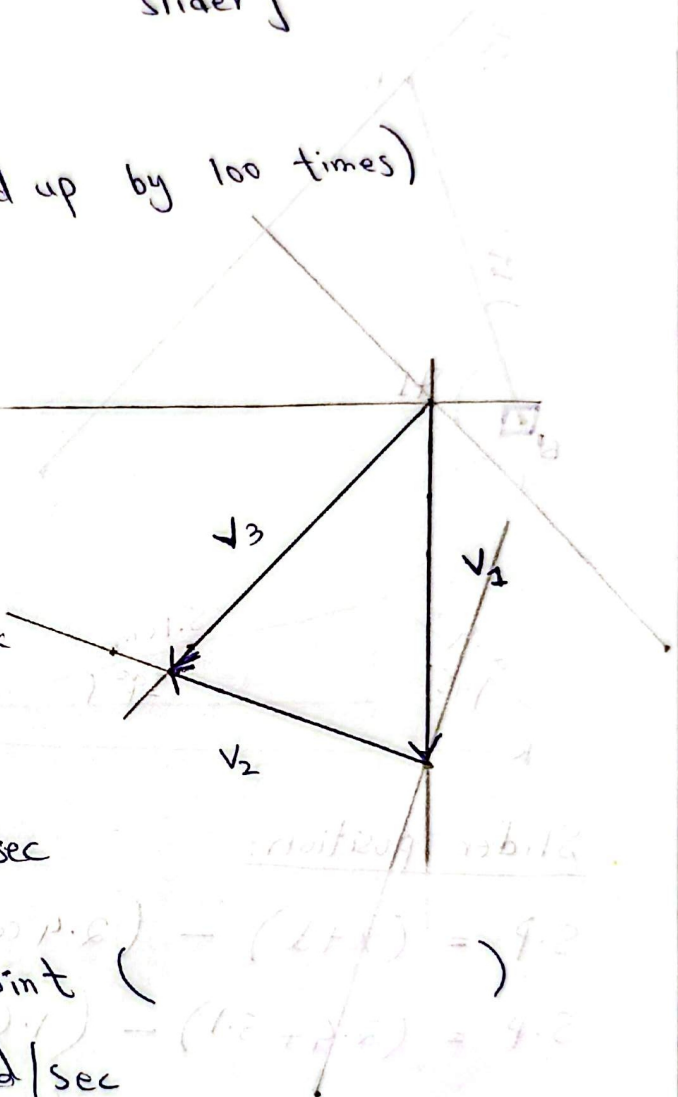
$$\omega_2 = \frac{0.039}{2.55} = 0.015 \text{ rad/sec}$$

$$\omega_3 = \frac{V_3}{r_3}$$

$$\omega_3 = \frac{0.0625}{1.2} = 0.052 \text{ rad/sec}$$

angular velocity at end point

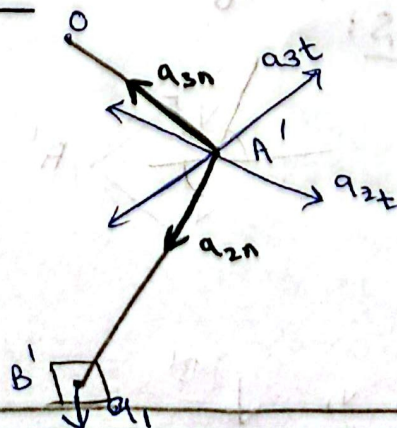
$$\omega_3 = \frac{0.0625}{5} = 0.0125 \text{ rad/sec}$$



Acceleration Analysis:

$$a_1 = 0 \text{ ft/sec}^2$$

because V_1 is constant



$$a_{2n} = r_2 \omega_2^2 = 2.55 (0.015)^2 = 0.00059 \text{ ft/sec}^2$$

$$a_{3n} = r_3 \omega_3^2 = 1.2 (0.052)^2 = 0.00325 \text{ ft/sec}^2$$

Scaling up 3000 times

$$a_{3n} = 0.00325 \times 3000 = 9.75 \text{ cm}$$

$$a_{2n} = 0.00059 \times 3000 = 1.77 \text{ cm}$$

$$a_{2t} = 10.8 \text{ cm} = 0.0036 \text{ ft/sec}^2$$

$$a_{3t} = 6.3 \text{ cm} = 0.0021 \text{ ft/sec}^2$$

$$\alpha_{2t} = \frac{a_{2t}}{r_2}$$

$$\alpha_{2t} = \frac{0.0036}{2.55}$$

$$\alpha_{2t} = 0.0014 \text{ rad/sec}^2$$

$$\alpha_3 = \frac{a_{3t}}{r_3}$$

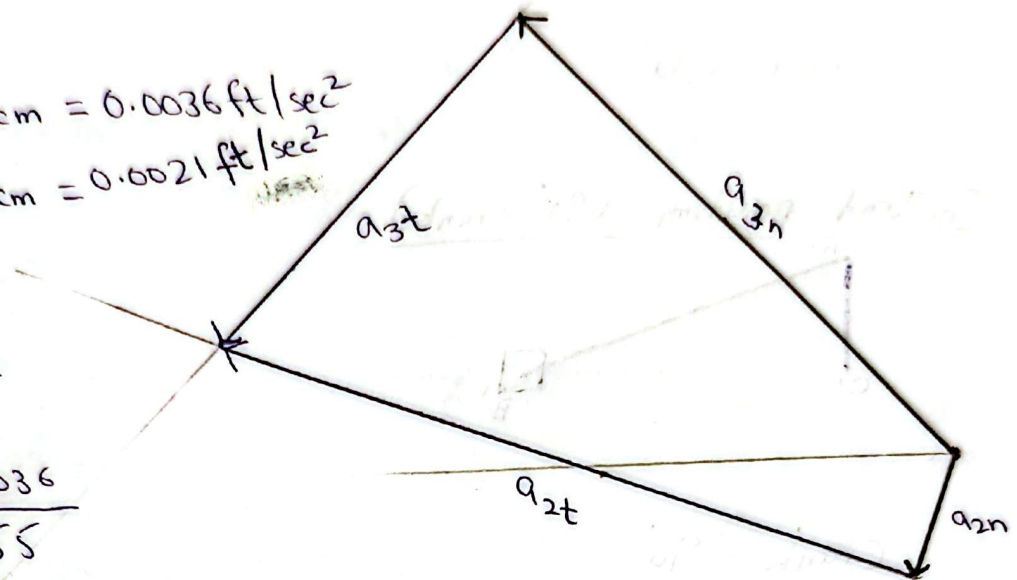
$$\alpha_3 = \frac{0.0021}{1.2} = 0.00175 \text{ rad/sec}^2$$

$$a_2 = \sqrt{(0.00059)^2 + (0.0036)^2}$$

$$a_2 = 0.003648 \text{ ft/sec}^2$$

$$a_3 = \sqrt{(0.00325)^2 + (0.0021)^2}$$

$$a_3 = 0.003869 \text{ ft/sec}^2$$



CRANK-SLIDER

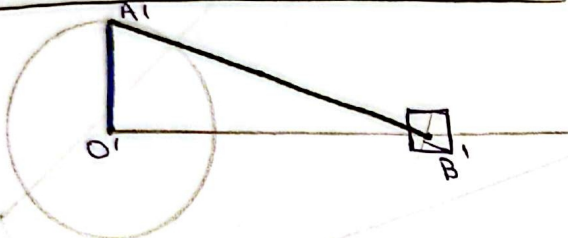
Extreme Position:



Crank = 0°

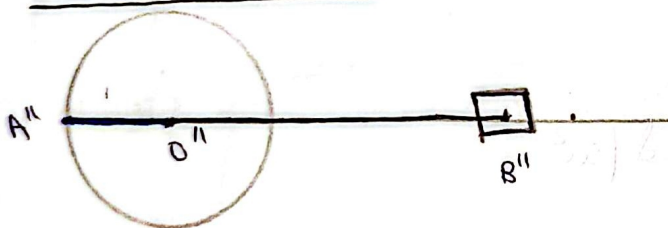
All of these cranks are vertical, they are just drawn horizontal nothing else.

Central position (90° crank)



Crank = 90°

Mean position:



Crank = 180°

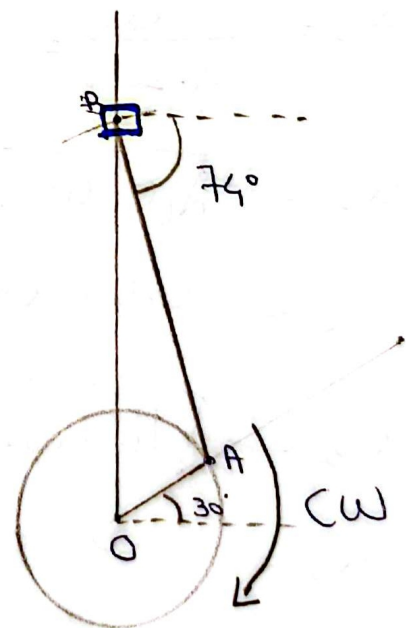
Position Analysis:

Coupler angle = 0

Slider $\rightarrow$ Extreme position

Crank = 0°

When crank is 60° , Coupler = 74°

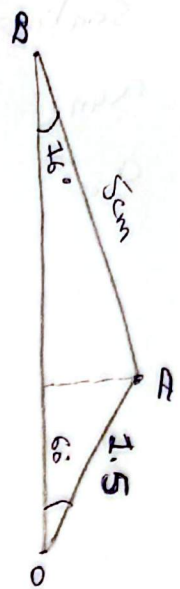


Slider position:

$$S.P = (OB)_{\text{Extreme}} - (OB)_{\text{at } 60^\circ}$$

$$S.P = (1.5 + 5) - (5 \cos 16^\circ + 1.5 \cos 60^\circ)$$

$$S.P = 0.944 \text{ cm}$$



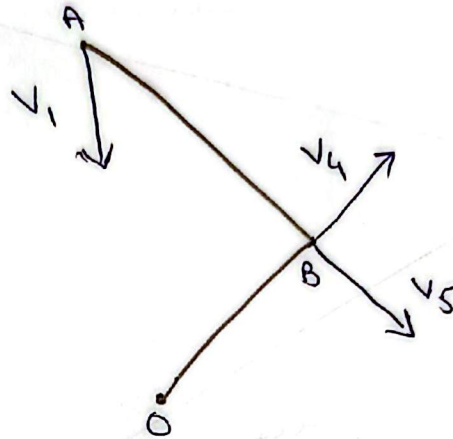
Velocity Analysis:

$$V_1 = 0.05 \text{ ft/sec}$$

Velocity Diagram:

Scaling up 100 times

$$V_1 = 0.05 \times 100 = 5 \text{ cm}$$

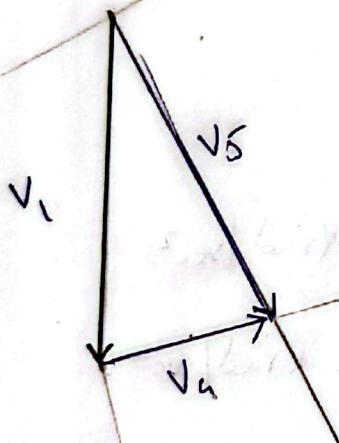


$$V_5 = 4.9 \text{ cm} = 0.049 \text{ ft/sec}$$

$$V_4 = 2.6 \text{ cm} = 0.026 \text{ ft/sec}$$

$$\omega_4 = \frac{0.026}{2.5} = 0.0104 \text{ rad/sec}$$

$$\omega_5 = \frac{0.049}{1.5} = 0.03267 \text{ rad/sec}$$



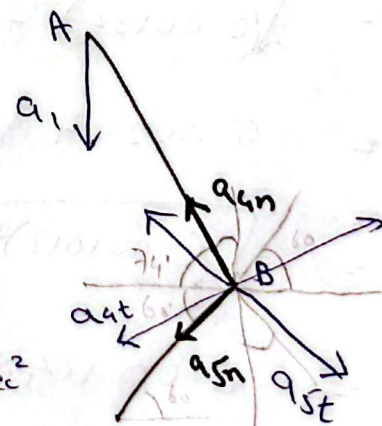
Acceleration analysis:

$$a_1 = 0$$

$\rightarrow V_1$ is constant

$$a_{4n} = \frac{(0.026)^2}{2.5} = 0.0002704 \text{ ft/sec}^2$$

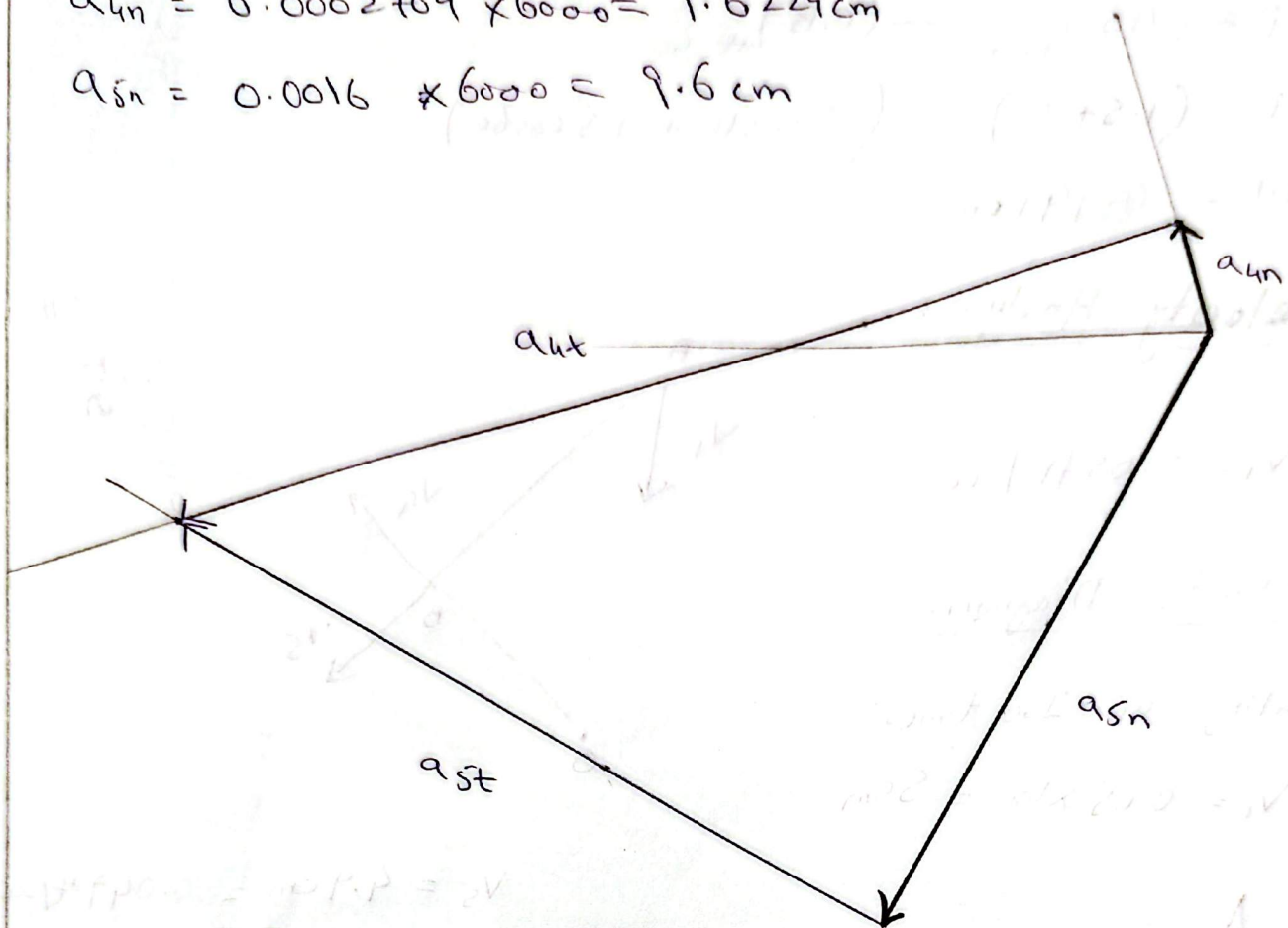
$$a_{5n} = \frac{(0.049)^2}{1.5} = 0.0016 \text{ ft/sec}^2$$



Scaling up 6000 times

$$a_{4n} = 0.0002704 \times 6000 = 1.6224 \text{ cm}$$

$$a_{5n} = 0.0016 \times 6000 = 9.6 \text{ cm}$$



$$a_{st} = 11.9 \text{ cm} = 0.00198 \text{ ft/sec}^2$$

$$a_{4t} = 15.2 \text{ cm} = 0.00253 \text{ ft/sec}^2$$

$$\alpha_s = \frac{a_{st}}{r_s} = \frac{(0.00198)}{1.5} = 0.00132 \text{ rad/sec}^2$$

$$\alpha_4 = \frac{a_{4t}}{r_4} = \frac{(0.00253)}{2.5} = 0.001012 \text{ rad/sec}^2$$

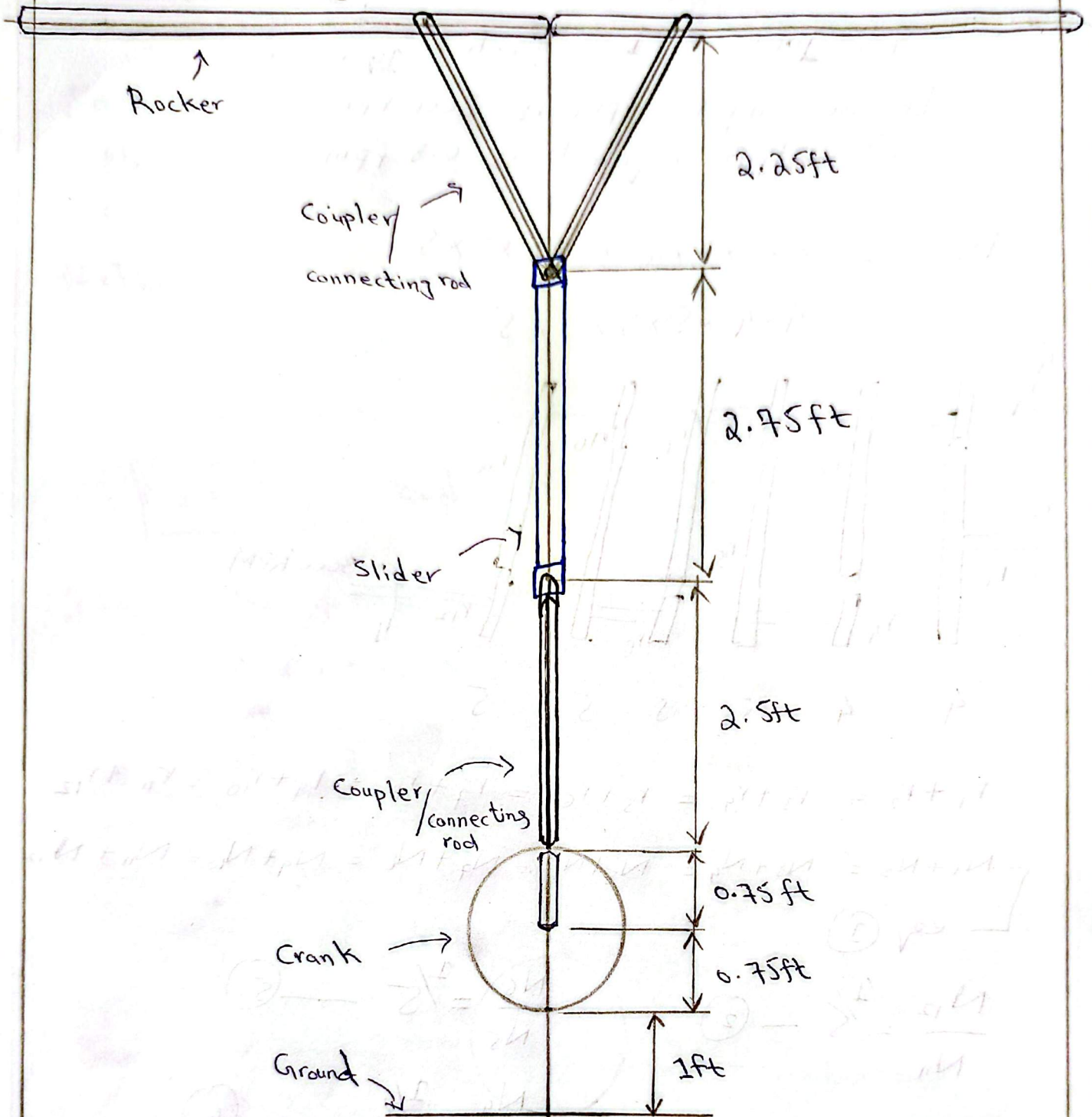
$$a_s = \sqrt{(0.00132)^2 + (0.0016)^2}$$

$$a_s = 0.00207 \text{ ft/sec}^2$$

$$a_4 = \sqrt{(0.001012)^2 + (0.0002704)^2}$$

$$a_4 = 0.001048 \text{ ft/sec}^2$$

Schematic Diagram:



Motor Type:

$$\left\{ \begin{aligned} \text{output RPM} &= \frac{\omega \times 60}{2\pi} = \frac{0.03267 \times 60}{2 \times \pi} \\ &= 0.3 \text{ rpm} \end{aligned} \right.$$

- AC Single-Phase Induction Motor with worm Gearbox
- Power 1/4hp
- RPM of motor is 3000 ← input rpm

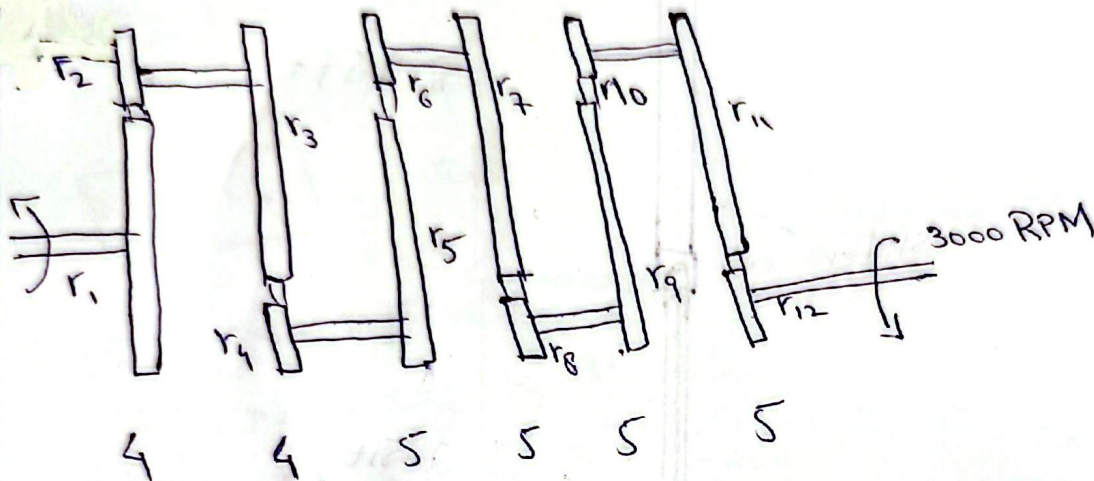
- Gear Box of Gear Ratio

$$1 : 10,000 = 1 \quad G.R = \frac{1}{10000}$$

because input rpm is 3000 rpm
and output required is 0.3 rpm

$$10000 = 2 \times 2 \times 2 \times 2 \times 5 \times 5 \times 5 \times 5$$

$$= 4 \times 4 \times 5 \times 5 \times 5 \times 5$$



$$r_1 + r_2 = r_3 + r_4 = r_5 + r_6 = r_7 + r_8 = r_9 + r_{10} = r_{11} + r_{12}$$

$$N_1 + N_2 = N_3 + N_4 = N_5 + N_6 = N_7 + N_8 = N_9 + N_{10} = N_{11} + N_{12}$$

eq (1)

$$\frac{N_{12}}{N_{11}} = \frac{1}{5} \quad \text{--- (2)}$$

$$\frac{N_6}{N_5} = \frac{2}{5} \quad \text{--- (5)}$$

$$\frac{N_4}{N_3} = \frac{2}{4} \quad \text{--- (6)}$$

$$\frac{N_{10}}{N_9} = \frac{2}{5} \quad \text{--- (3)}$$

$$\frac{N_8}{N_7} = \frac{2}{5} \quad \text{--- (4)}$$

$$\frac{N_2}{N_1} = \frac{2}{4} \quad \text{--- (7)}$$

$$N_{11} = 5 N_{12}$$

$$N_5 = 5 N_6$$

$$N_9 = 5N_{10}$$

$$N_3 = 4N_4$$

$$N_1 = 4N_2$$

$$N_7 = 5N_8$$

So

$$N_1 + N_2 = N_3 + N_4$$

$$(\text{assume } N_1 = 16)$$

$$(4 \times 16) + 16 = N_3 + N_4$$

$$\text{Put } N_3 = 4N_4$$

$$80 = 5N_4$$

$$\boxed{N_4 = 16}$$

and

$$N_3 = 4 \times 16 = 64$$

$$\text{Now } N_9 + N_{10} = N_3 + N_4$$

$$N_9 + N_{10} = 16 + 64$$

$$N_9 + N_{10} = 80$$

$$5N_{10} + N_{10} = 80$$

$$N_{10} = \frac{80}{6} = \frac{40}{3}$$

$$\text{Same } N_9 = \frac{5 \times 40}{3} = \frac{200}{3}$$

$$\text{Same } N_7 = \frac{200}{3} \quad N_5 = \frac{200}{3}$$

$$N_8 = \frac{40}{3} \quad N_6 = \frac{40}{3}$$

$$N_{11} = \frac{40}{3}$$

$$N_{12} = \frac{200}{3}$$

Sources of Errors:

- ⇒ Inaccuracy in drawing link lengths during graphical synthesis of slider crank mechanism.
- ⇒ Errors in making crank radius, coupler lengths & slider travel distance on drawing sheet.
- ⇒ Parallax error while reading measuring instruments - if not viewed perpendicularly.
- ⇒ Human error during manual marking, cutting & drilling of links and joints.
- ⇒ Misalignment between pivot pin holes causing the mechanism to bind or wobble.

Conclusion:

This project represents a functional and well-referenced design for an umbrella opening and closing mechanism based on a slider-crank linkage system. Inspired by the iconic retractable umbrella structures of Masjid al-Nabwi in Al-Madinah, the mechanism demonstrates how sophisticated large-scale engineering principles can be scaled down for academic study and practical application. Its application effectively converts the rotary output of a electric gear motor into linear motion required to actuate the arms.

Future improvements may include solar panels for autonomous operation, wind load sensors for storm protection, aligning with the vision demonstrated by the world-renowned Madinah Umbrella system.

$$N_1 = 64 \times 3 = 192$$

$$N_2 = 16 \times 3 = 48$$

$$N_3 = 64 \times 3 = 192$$

$$N_4 = 16 \times 3 = 48$$

$$N_5 = \frac{200}{3} \times 3 = 200$$

$$N_6 = \frac{40}{3} \times 3 = 40$$

$$N_7 = \frac{200}{3} \times 3 = 200$$

$$N_8 = \frac{40}{3} \times 3 = 40$$

$$N_9 = \frac{200}{3} \times 3 = 200$$

$$N_{10} = \frac{40}{3} \times 3 = 40$$

$$N_{11} = \frac{200}{3} \times 3 = 200$$

$$N_{12} = \frac{40}{3} \times 3 = 40$$

↳ Each multiplied by 3

